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KAPASA MAKASA UNIVERSITY

SCHOOL OF APPLIED SCIENCES AND OPEN LEARNING

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

ACCESS CONTROL AND INTRUSION DETECTION SYSTEMS ICT 361

DURATION: 3 HRS DATE:28th October, 2025

Total Marks: 100

General Instructions to Candidates:

1. Answer questions as instructed.
2. No unauthorized materials are allowed
3. No electronic devices are allowed.
4. Start **each** answer on a new page.
5. Write clearly and legibly.

Exam Instructions:

This question paper contains two sections; SECTION A and SECTION B. Answer all questions in SECTION A and answer two questions from SECTION B. Note that question one in SECTION B is Compulsory.

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SECTION A [Answer all questions in this section – 50 Marks]

(5) 1. Define the terms identification, authentication, and authorization in access control, and explain how they relate to one another. [5 marks] ✓

(5) 2. Differentiate between Mandatory Access Control (MAC) and Discretionary Access Control (DAC), giving one example of where each might be applied. [5 marks] ✓

(3) 3. Explain the principle of 'Least Privilege' and describe why it is considered a best practice in access control. [5 marks] ✓

(3) 4. Briefly describe the purpose of Access Control Lists (ACLs) and mention one limitation of this approach. [5 marks] ✓

(5) 5. Differentiate between Anomaly Detection and Misuse Detection techniques in Intrusion Detection Systems. Provide one advantage and one limitation of each. [5 marks] ✓

(5) 6. Explain the differences between Host-based Intrusion Detection Systems (HIDS) and Network-based Intrusion Detection Systems (NIDS). Include one advantage and one disadvantage of each approach. [5 marks] ✓

(5) 7. Describe the role of an Intrusion Prevention System (IPS). How does it differ from an Intrusion Detection System (IDS)? [5 marks] ✓

(5) 8. Briefly discuss the purpose and function of the Snort Intrusion Detection System, highlighting at least three of its main features. [5 marks] ✓

(5) 9. What is a vulnerability assessment in the context of intrusion detection and prevention? Mention two types of vulnerability scans and their purposes. [5 marks] ✓

(3) 10. Explain the term "Hybrid Intrusion Detection System" and discuss how it combines the advantages of different IDS approaches. [5 marks] ✓

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SECTION B (They are three questions in this section choose only two. Question 1 is Compulsory. – 50 Marks)

1. ZedNet, a mid-sized Internet Service Provider (ISP), recently experienced a sophisticated cyberattack. Attackers used a Distributed Denial of Service (DDoS) attack and later attempted to install malware through phishing emails. Although ZedNet has a basic firewall in place, the attack was only detected after customers began reporting service disruptions. To strengthen their cybersecurity posture, the company is considering deploying an Intrusion Detection and Prevention System (IDPS) to monitor traffic, detect malicious activities, and prevent future attacks.

Questions:

a) Differentiate between **Network-based IDPS (NIDPS)** and **Host-based IDPS (HIDPS)**. Which would be more suitable for ZedNet's situation? Justify your answer. [6 marks]

b) Describe **three types of attacks** that an IDPS can detect and prevent, referencing how these relate to ZedNet's incident. [6 marks]

c) Explain the difference between **signature-based detection** and **anomaly-based detection**, and discuss the advantages and disadvantages of each. [7 marks]

d) Recommend a deployment strategy for ZedNet to implement IDPS, including key steps to ensure ongoing monitoring, updates, and response to alerts. [6 marks]

2. Mulakupikwa Bank is a growing financial institution that recently migrated its services to a hybrid cloud environment. The organization handles sensitive customer data, including financial transactions and personally identifiable information (PII). Currently, access to internal systems is managed through a simple username-password system.

Recently, an internal audit revealed multiple security gaps:

- Employees share login credentials to speed up work.
- Contractors have the same level of access as permanent staff.
- Remote workers access the network using unsecured devices.
- No centralized logging exists to monitor access attempts.

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To improve security, management wants to implement a role-based access control (RBAC) system, enforce least privilege, and introduce multi-factor authentication (MFA) across all systems.

Questions:

a Identify and discuss **three** key weaknesses in the current access control setup. [6 marks]

b) Explain how Role-Based Access Control (RBAC) can help mitigate these weaknesses. Provide two practical examples of roles and their associated permissions in this organization. [7 marks]

c) Discuss how Multi-Factor Authentication (MFA) enhances security in this scenario. Include at least two MFA methods that would be suitable. [6 marks]

d) Propose an implementation plan (steps and considerations) for deploying RBAC and MFA in Mulakupikwa Bank, highlighting any potential challenges. [6 marks]

(5.1) (5.1) (4.0) (2.0)

3. Discuss the four major access control models—**Mandatory**, **Discretionary**, **Role-Based**, and **Rule-Based** Access Control. Compare their strengths, weaknesses, and typical applications in real-world systems. [25 marks]

5.1
5.1
4.0
2.0
16.2

16.2/25

25°/4 = 5.124

4
8
12
16
20
24
28
32
36
40
44
48

76
3
228

SECTION A = 44.0
SECTION B = 16.0
16.2
76.2%

76/100 × 80 = 228/5 = 45.6 = 46

EXAM = 46
CA = 37
83%

GRADE = A